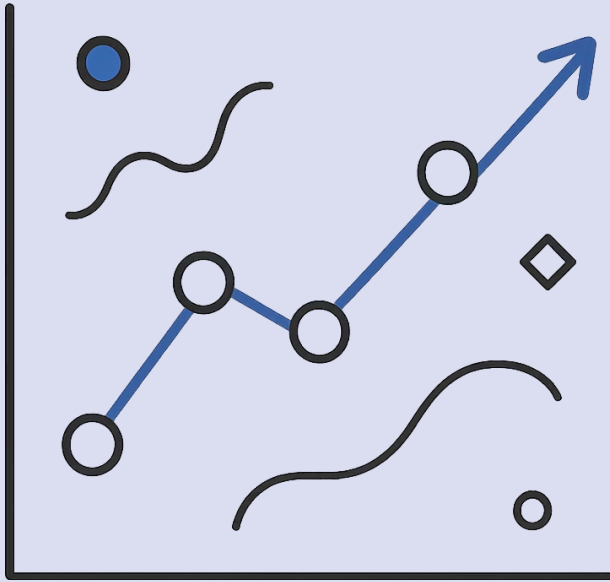




Simple Linear Regression

The simplest regression algorithm — used to predict one numerical output using one input feature via a straight-line relationship.

Why "Simple"?



It is called Simple because it uses only one input feature.

Input Feature

Area (sqft)

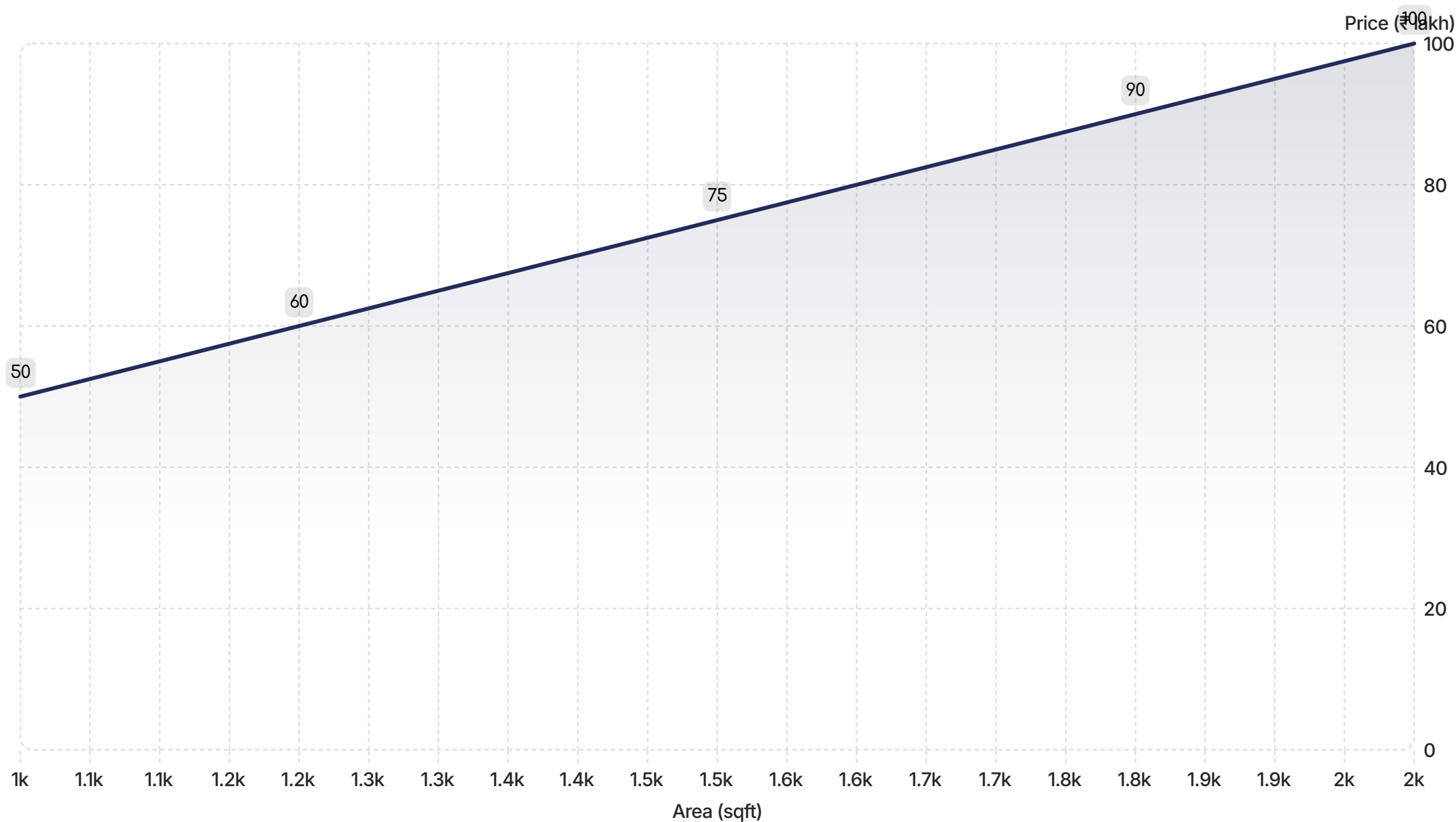
Target Variable

House Price

One feature → Simple Linear Regression.

Why "Linear"?

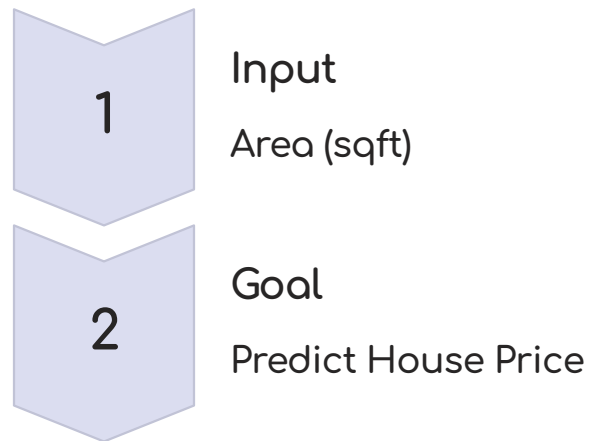
It is called Linear because the model draws a straight line through the data. As area increases, price increases proportionally.





Running Example

House price can depend on many features — area, bedrooms, age, distance from city. But for Simple Linear Regression, we use only one:



Example Data

Area (sqft)	Price (₹ lakh)
1000	50
1200	60
1500	75
1800	90
2000	100

This is our training dataset. The model learns the relationship between area and price from these five examples.

 Price is in lakhs (₹).



Human Thinking

If asked: Area = 1600 sqft $\rightarrow$ Price = ?

1500 sqft

₹75 lakh

1600 sqft

≈ ₹80 lakh

1800 sqft

₹90 lakh

This intuitive interpolation is exactly the pattern Simple Linear Regression learns.

Formula of Simple Linear Regression

$$y = mx + c$$

1

x
Input feature

2

y
Predicted output

3

m
Slope

4

c
Intercept

In ML terms: Predicted Price = slope × Area + intercept

Formula for House Price

$$Price = m \times Area + c$$

- ✓ Example learned formula:
Price = $0.05 \times \text{Area} + 5$

This means the model has learned a straight-line relationship between area and price.

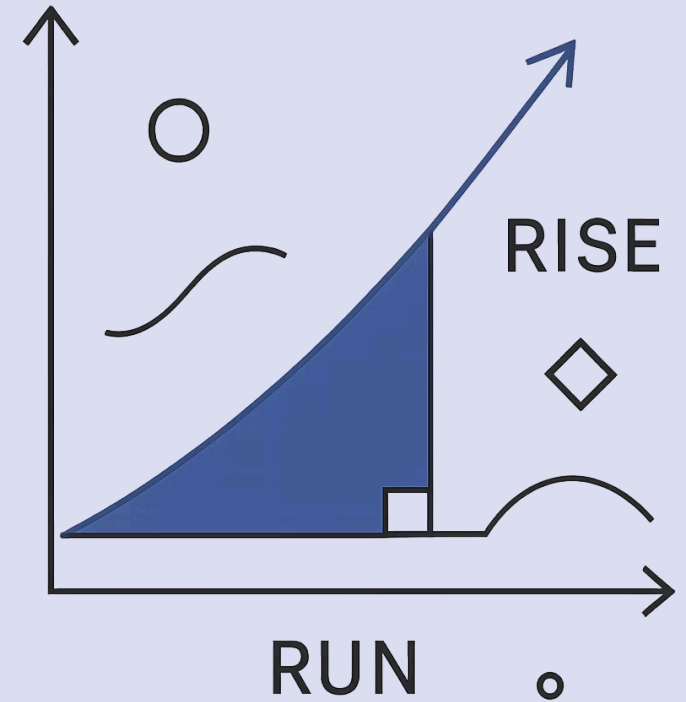
Plug in any area value and the formula returns a predicted price.

What Is Slope?

Slope = how much y changes when x increases by 1 unit.

In Our Example

Price = $0.05 \times \text{Area}$ → every extra sqft adds ₹5,000 (0.05 lakh).



What Is Intercept?

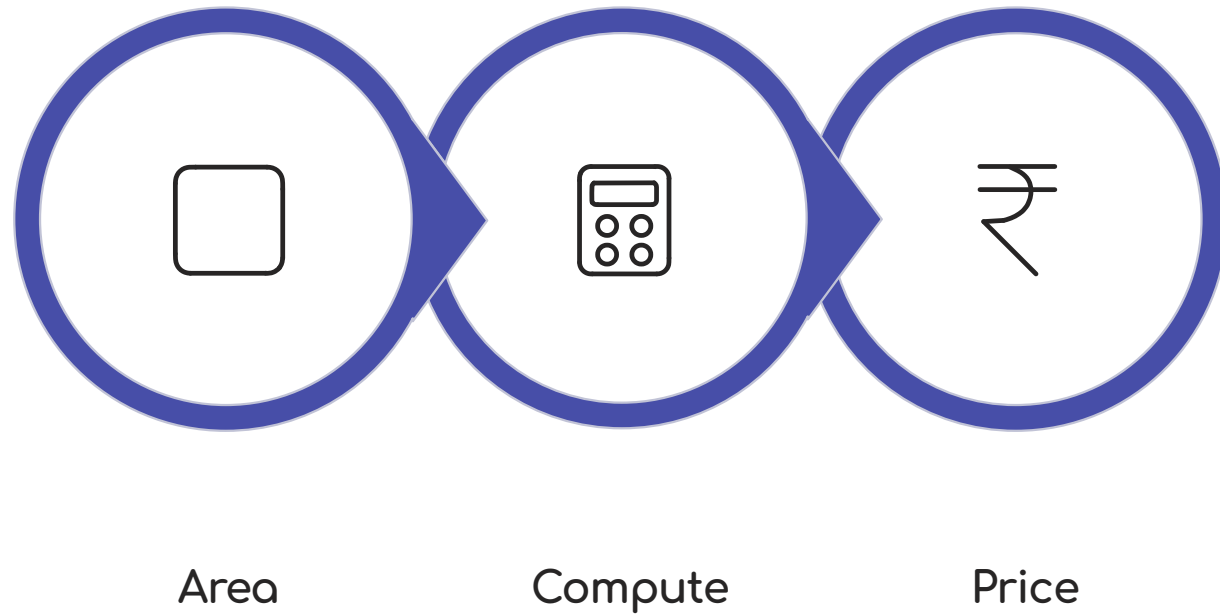
$$Price = 0.05 \times Area + \underbrace{5}_{\text{intercept}}$$

Intercept is the starting/base value of the line — it helps place the line correctly on the graph.

□ Intercept = 5 in this example.

Example Calculation

Model learned: $\text{Price} = 0.05 \times \text{Area} + 5$



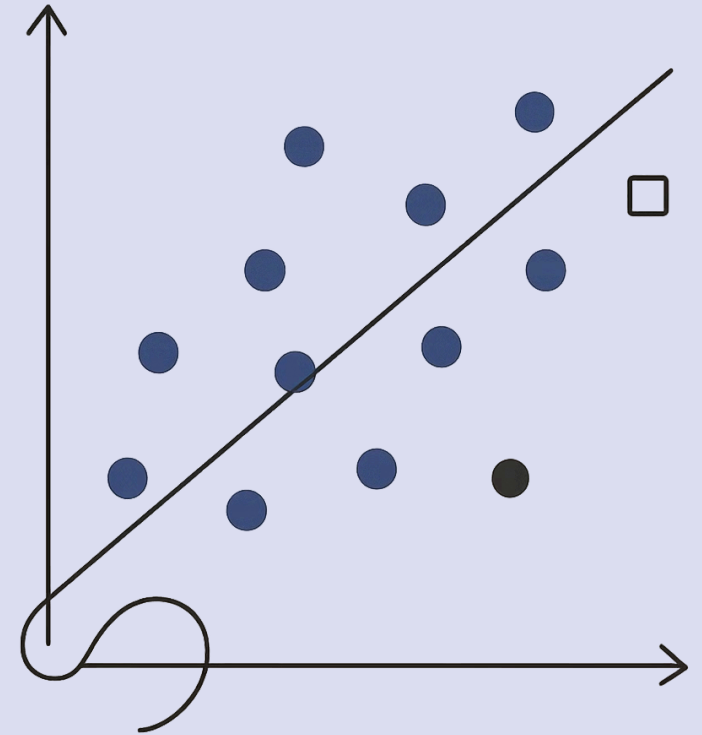
Plugging in 1600 sqft gives a predicted price of ₹85 lakh.

How Does the Model Learn?

The model looks at all training data and draws the best possible straight line — passing close to most points, but not necessarily through every one.

Noise Sources

Location, interior quality, road access, parking, builder reputation, market conditions



Prediction Error

Actual Price

₹82 lakh

Predicted Price

₹85 lakh

Error

₹3 lakh

Simple Linear Regression tries to find a line where the total error across all points is as small as possible.



Some error is normal — real data has noise.

Visual Understanding



Data Points

Each house shown as a dot



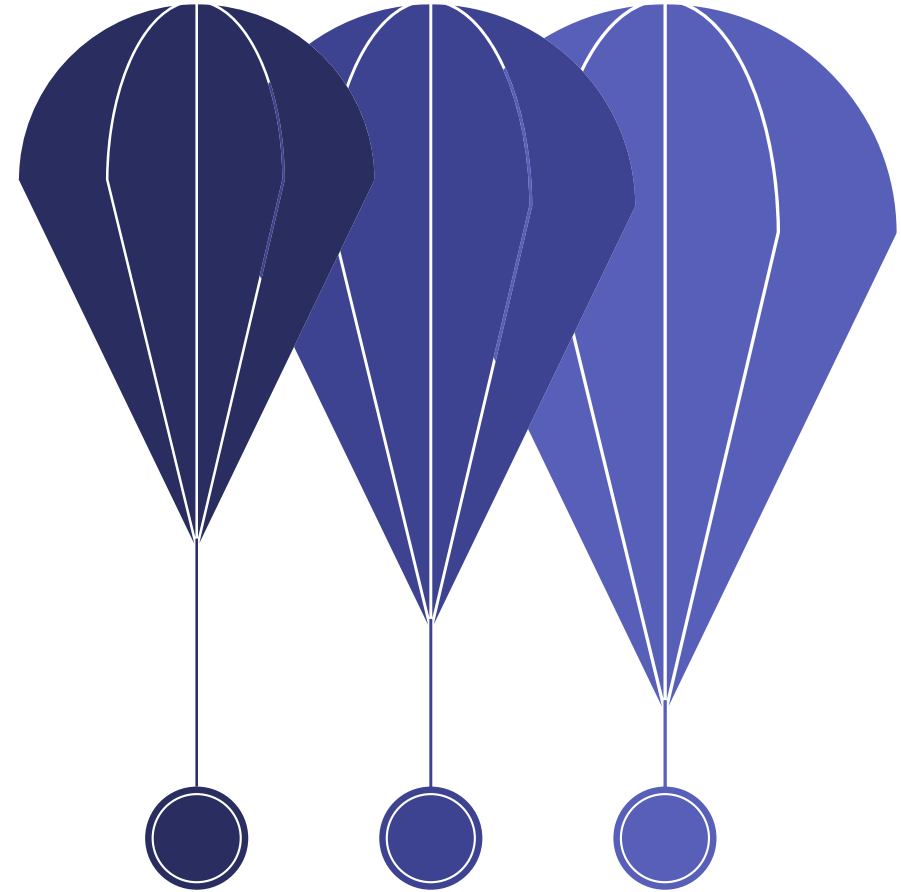
Trend Line

A straight line fits near the dots



Predict Price

New point on the line gives prediction



Each house is a dot. The model draws a line near the dots. For a new area, it reads the line to predict price.

Python Step 1: Import Libraries

```
import pandas as pd  
from sklearn.linear_model import LinearRegression
```

pandas

Used for data tables (DataFrames).

LinearRegression

The ML model from scikit-learn.

Python Step 2: Create Data

```
data = {  
    "area": [1000, 1200, 1500, 1800, 2000],  
    "price": [50, 60, 75, 90, 100]  
}  
df = pd.DataFrame(data)  
print(df)
```

This creates our small training dataset as a pandas DataFrame.

Python Step 3: Separate X and y

```
X = df[["area"]]  
y = df["price"]
```

X

Input feature (2D table — double brackets required by scikit-learn)

y

Target variable (1D series)

Python Steps 4 & 5: Create & Train

```
model = LinearRegression() # Step 4: Create  
model.fit(X, y)           # Step 5: Train
```

1

Create

Empty model — nothing learned yet

2

Train

Model learns area → price relationship

Python Step 6: Predict

```
new_house = pd.DataFrame({"area": [1600]})  
predicted_price = model.predict(new_house)  
print("Predicted price:", predicted_price[0], "lakhs")
```

✓ Output: Predicted price: 80.0 lakhs

Full Code

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data = {"area": [1000, 1200, 1500, 1800, 2000],
        "price": [50, 60, 75, 90, 100]}
df = pd.DataFrame(data)

X = df[["area"]]
y = df["price"]

model = LinearRegression()
model.fit(X, y)

new_house = pd.DataFrame({"area": [1600]})
predicted_price = model.predict(new_house)
print("Predicted price:", predicted_price[0], "lakhs")
```

Checking Slope & Intercept

```
print("Slope:", model.coef_[0])  
print("Intercept:", model.intercept_)
```

✓ Example output:
Slope: 0.05
Intercept: 0.0

This confirms the model learned:

$$Price = 0.05 \times Area + 0$$

Important Limitation

Simple Linear Regression uses only one feature. But real house price depends on many things:

Location

Bedrooms

House Age

Parking

Metro Access

Build Quality

Simple Linear Regression is great for learning, but real projects usually need more features.

When to Use Simple Linear Regression

Use it when there is one main input, a numerical output, and a roughly straight-line relationship.



Study Hours → Marks



Experience → Salary

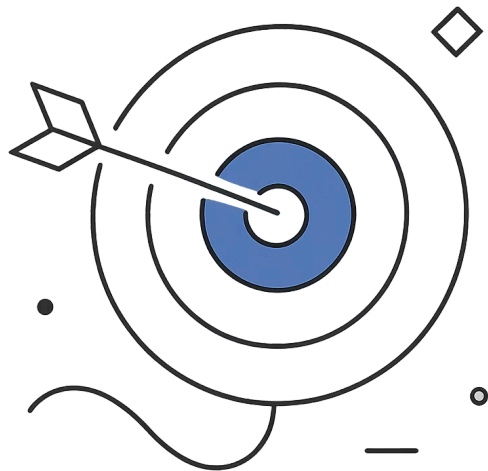


Ad Spend → Sales



Temperature → Ice Cream Sales

Common Beginner Mistake #1



Thinking Prediction Is Always Exact

Regression gives an estimate, not a guarantee.

Predicted

₹80 lakh

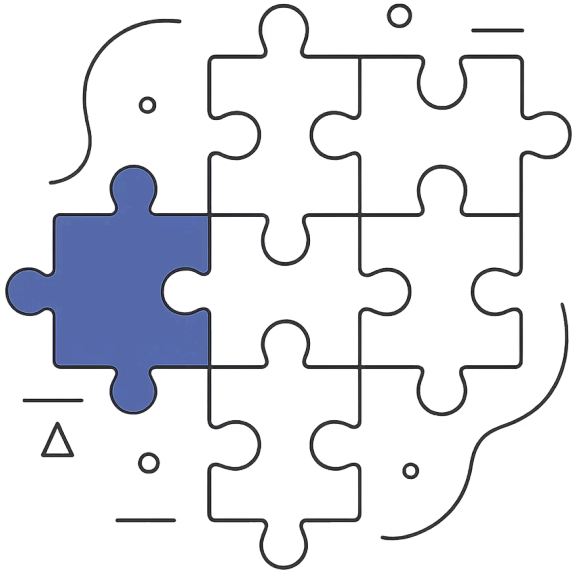
Actual

₹83 lakh



A small difference is completely normal.

Common Beginner Mistake #2



Using Simple Linear Regression for Every Problem

Simple Linear Regression uses only one input.

Real-world problems usually need multiple inputs — that's where Multiple Linear Regression comes in.

Common Beginner Mistake #3

Confusing Slope and Intercept

Slope

Rate of change

Price increase per sqft

Intercept

Starting/base value

Where the line begins

❏ For house price: Slope = ₹ added per sqft. Intercept = line starting point.

Final Summary

Simple

One input feature

Linear

Straight-line relationship

Regression

Predicts a number

Area → Price | Formula: $\text{Price} = \text{slope} \times \text{Area} + \text{intercept}$

Memory Line

Simple. Linear. Regression.

01

Simple

One input

02

Linear

Straight line

03

Regression

Predicts a number

Up Next

Multiple Linear Regression

Now that you understand one input, it's time to scale up — using multiple features to make even better predictions.

